

Unity announces support for open source robotics middleware suite

Unity, the creator and operator of real-time 3D (RT3D) content, has announced support for ROS 2 – the open source robotics middleware suite from Open Robotics. Building on its support of ROS earlier this year, Unity is now supporting ROS 2 due to significant advancements and support of more hardware drivers, networking modules, communication architecture, and several robot algorithms.

“The addition of real-time support makes it suitable for time-critical and high-performance systems, including AMRs, which we demonstrate with Unity running a ROS 2 navigation stack,” said Dr Danny Lange, SVP of AI and machine learning, Unity.

The company has also released a new sample AMR project. The demo showcases a simple warehouse environment, a fully articulated model of a Turtlebot 3 mobile robot with simulated LIDAR and motor controllers, and a Dockerfile to make it easy to build an image containing all of the ROS 2 dependencies necessary to exercise the mapping and navigation stacks against simulation.



“This demo is another powerful example of how we take a flexible framework for writing robot software, and couple it with our robotics packages so that users can more easily create intelligent robots,” said Lange.

“Simulation is a critical part of developing robust robot systems and it’s important for developers to be able to choose the right simulator for their application. It is a welcome addition to the growing community of roboticists who are using open source software to accelerate the development of advanced systems,” he added.

While it is actually hard to evolve an operating system kernel, the rate of innovation at this level has been lower than the functionality implemented outside of the system. The eBPF is said to change this mechanism fundamentally, by permitting to run sandboxed programs with the operating system. Application



developers can run eBPF programs to add capabilities to the operating system at runtime.

The operating system then guarantees safety and execution efficiency, says the eBPF Foundation spokesperson. It allows users to even combine and apply

logic across multiple subsystems which were traditionally completely independent.

The Foundation is expected to be the home for open source eBPF projects and technologies to drive the growth and adoption of the eBPF ecosystem.

The eBPF Foundation will be hosted by the Linux Foundation.

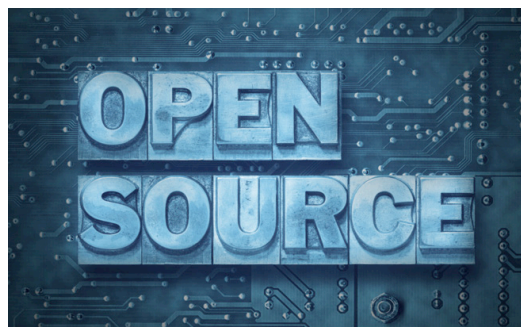
Study finds vulnerabilities in open source components used in COTS

A recent study has found critical vulnerabilities in the commonly used open source components in commercial off-the-shelf (COTS) IT products.

The white paper by GrammaTech and Osterman Research found a wide-ranging use of open source components in the COTS products, and said that meeting and email client COTS products are the most vulnerable.

“Many open source components contain a range of known vulnerabilities that can be used as egress points for cyberattacks. Lack of awareness of open source components used by organisations in commercial off-the-shelf software increases the security risk, attack surface, and potential for compromise by cybercriminals,” reads the study.

Among the applications analysed in groups of five, 30 per cent of all the open source components contained at least one vulnerability that has already been



identified as a common vulnerability and exposure (CVE). The email and meeting tools have the “highest average weighting” of vulnerabilities, which is concerning given their widespread use across organisations.

Of the components identified across the

applications analysed by CodeSentry and used in this study, two versions of the Firefox open source component (not the browser itself) contributed 75.8 per cent of the CVEs. In second place, 16 versions of OpenSSL had a combined 9.6 per cent of the CVEs, and two versions of Libav comprised 8.3 per cent of the CVEs. These numbers are derived by counting the number of vulnerabilities in each component when the latter is used in an application.

Acquia renews support for the Drupal Steward Program

Digital experience company Acquia has announced that it is renewing its founding partnership support of the Drupal Steward Program, a Web application firewall introduced by the Drupal Association and operated jointly with the Drupal security